Field Hours

CSCE 748 Computational Photography

Original notes for this site. Same topics as the catalog course. Not a reprint of another university's lectures, and not academic credit.

Lecture 1 · 90 minutes

Time, aperture, and a disk

You form a photograph with a lens that gathers a bundle of rays. A point at 2 meters lands on one sensor cell. A point at 5 meters lands as a disk, the defocus blur. The sensor integrates that disk for 0.01 seconds, and a moving subject also smears. You have two blurs, depth and motion, in the same exposure. Naming which blur you have is the first measurement.

You set exposure as a product you can double in more than one way. Doubling the time from 0.01 seconds to 0.02 seconds doubles the collected energy and can saturate a cell. Opening the aperture by one stop also doubles the energy, and it changes the defocus disk, because a wider aperture takes a wider bundle from the point at 5 meters. A brightness slider after the fact is a later multiply. It is not this integral on the sensor.

Focus picks which distance is a point. Aperture and time both change energy.

Point at 2 meters

Lands on one cell

Time is 0.01 seconds

Motion can still smear

Point at 5 meters

Lands as a disk

Wider aperture, larger disk

One stop doubles energy

Two ways to double the energy

1. A point at 2 meters is sharp on one cell. A point at 5 meters is a defocus disk.

2. Time of 0.01 seconds collects some energy. Time of 0.02 seconds collects twice as much, and a cell may saturate.

3. One stop wider also doubles the energy, and it enlarges the disk from the point at 5 meters.

Problem 1. You double the time from 0.01 seconds to 0.02 seconds and leave the aperture fixed. What happens to the collected energy, and what happens to defocus size?

Answer. The energy doubles, so a cell near the top of its range can saturate. The defocus size does not change, because the aperture stayed fixed.

Problem 2. You open the aperture by one stop and keep the time at 0.01 seconds. What two things change?

Answer. The collected energy doubles, and the defocus disk of a point at 5 meters grows. A point at the 2 meter focus can stay on one cell.

Lecture 2 · 90 minutes

A kernel after the shutter

You filter the grid with a kernel chosen for a photographic reason. A 3 by 3 box of weights 1/9 cuts noise and also widens the defocus disk from the point at 5 meters. Use it only when noise is the larger problem. The 9 weights must add to 1. If each weight is 1, a flat gray of 20 becomes 180, an exposure change hiding in the filter.

You sharpen only where the sensor still has range. Subtract a blur from the original and add a fraction of the difference. A pixel at 80, a blur at 50, and a fraction of 0.5 give 95. You do not sharpen a pixel already at 255. There is no detail above the clip, and the difference would invent a halo. Leave 255, and sharpen neighbors that still have room.

A blur whose weights add to 1 keeps a flat gray. Sharpen skips a clipped pixel.

Weights must sum to 1 -> Blur into a new buffer -> Add a fraction back -> Leave 255 alone

A pixel at 80

1. The blur at that pixel is 50. The difference is 30.

2. Half of 30 is 15. The sharpened value is 95.

3. A neighbor already at 255 stays 255. You do not build a halo from a difference the clip cannot support.

Problem 1. Each of 9 kernel weights is 1, and the pixel sits in a flat region of value 20. What is the output, and what is wrong?

Answer. The output is 180, because 9 times 20 is 180. The weights add to 9 rather than 1, so the filter changes brightness. A box blur uses 1/9 for each weight.

Problem 2. A pixel is 80, its blur is 50, and you add half the difference. What do you store, and what would you refuse if the pixel were 255?

Answer. You store 95, because half of 30 is 15 and 80 plus 15 is 95. If the pixel were already 255 you would leave it, because there is no detail above the clip.

Lecture 3 · 90 minutes

Four pairs and a seam

You stitch two views with a homography when the scene is far, or the camera rotated about its center. You mark 4 point pairs on the shared wall, with no 3 points on one line. Those pairs fix the map from view A to view B. You warp A with the inverse map so each output pixel pulls a sample. Fewer than 4 pairs leaves the map unfixed. A disagreeing fifth pair means the scene is not one plane.

You blend the overlap after the warp. If the overlap is 40 pixels wide, a pixel 10 pixels in takes weight 10/40 from the incoming view and 30/40 from the other. You blend the aligned images. Blending before the warp smears the seam in the wrong coordinate. A doubled parallax edge will not be fixed by that 40 pixel ramp. It needs a model other than one plane.

Four pairs fix the map. The ramp only hides a seam that is already aligned.

1. Mark 4 point pairs

2. Solve the homography

3. Warp by the inverse

4. Blend across 40 pixels

Ten pixels into a seam

1. Four pairs, no three on one line, fix the map from view A onto view B.

2. You warp A by pulling a sample for every output pixel. The overlap is 40 pixels wide.

3. A pixel 10 pixels into the overlap uses weight 0.25 from the incoming view and 0.75 from the other, because 10/40 is 0.25.

Problem 1. You have 3 point pairs. Can they fix a homography between the two views?

Answer. No. You need 4 pairs, and no three of the points may lie on one line. Three pairs leave the map underdetermined.

Problem 2. The overlap is 40 pixels wide and the pixel is 10 pixels into it. What is the incoming view weight?

Answer. 0.25. That is 10 divided by 40. The other view receives the remaining 0.75.

Lecture 4 · 90 minutes

Three exposures

You merge several exposures because one frame cannot hold both a lamp and a shadow. You shoot 3 frames at 0.01, 0.04, and 0.16 seconds. Each step multiplies time by 4. A pixel at 255 in the long frame is saturated, so you ignore it. The same pixel may read 40 in the short frame. You divide each usable sample by its time so the frames share one radiance unit.

You check linearity where a pixel is valid in two frames. At 0.01 seconds the sample is 20. At 0.04 seconds a linear sensor should read near 80, because the time grew by 4. A reading of 60 means a raw divide will bias the radiance. After the merge you tone-map into a display range of 0 to 255. That map is a display compress, not a clip of the long frame.

Drop clipped samples, scale the rest by time, then compress for display.

Three exposure times -> Drop samples at 255 -> Divide by the time -> Tone-map into 0 to 255

One pixel in two frames

1. The times are 0.01, 0.04, and 0.16 seconds. The long frame reads 255, so you drop that sample.

2. The short frame reads 20. If the sensor is linear, the middle frame should read near 80, because 4 times 20 is 80.

3. A middle reading of 60 means you cannot trust a raw divide. Usable samples are divided by their own times before the tone map.

Problem 1. The exposure times are 0.01, 0.04, and 0.16 seconds. By what factor does each step increase the time?

Answer. 4. You get 0.04 by multiplying 0.01 by 4, and 0.16 by multiplying 0.04 by 4.

Problem 2. A pixel is 255 in the longest frame and 40 in the shortest. Which sample enters the radiance estimate?

Answer. The short-frame sample, 40, divided by its exposure time of 0.01. The value 255 is saturated, so you do not average it in.

Lecture 5 · 90 minutes

Sixteen rays, not a sum

You store a light field when one summed image is not enough to refocus later. Each sample records a ray by where it hits a lens plane and where it hits a sensor plane. This exercise keeps 4 viewpoints and 4 pixels, which is 16 rays, not a finished photograph. An ordinary exposure would sum every lens-plane sample that lands on one sensor cell. You refuse that sum at capture so you can choose a different sum after the shutter.

You refocus by shifting, then adding. For the point at 2 meters, you average the 4 viewpoint samples at shift 0. To focus closer, you shift a neighbor view by 1 pixel before the add. A point at 5 meters that needed a different shift stays a disk. You did not delete defocus. You chose which of the two depths the average makes sharp.

Keep the rays apart. A shift chooses the depth that the average will sharpen.

4 views and 4 pixels

16 rays, not summed

Shift by 0 or by 1

Average the 4 views

A shift of one pixel

1. You stored 4 views instead of the lens sum. The toy grid holds 16 rays.

2. To focus at 2 meters you add the 4 view samples with shift 0 and average them.

3. To focus closer you shift neighbor views by 1 pixel, then average. The point at 5 meters stays soft if it needed another shift.

Problem 1. You stored 4 viewpoints and 4 pixels. How many rays is that, and why not store the summed photo?

Answer. 16 rays. The summed photo would throw away the lens-plane split, so a later shift could not refocus.

Problem 2. The point at 2 meters is sharp at shift 0. What do you do with the 4 view samples?

Answer. Add them with no shift, then average. A shift of 1 pixel is the choice for a closer depth, and it would leave the 2 meter point less sharp.

Lecture 6 · 90 minutes

A cheap path down the image

You change width by removing seams, not by scaling every column. On this 8 by 8 image a seam is 8 pixels, one per row, each step touching a neighbor. You score a pixel by its difference from its horizontal neighbors. You remove the path whose 8 scores sum to the least. A uniform scale would shrink a face and a blank wall together. The seam tries to take the wall.

You protect a region with a penalty. The face occupies columns 3 through 5, and you add 100 to every score there. A path through the face then loses unless the wall costs more than 100 per pixel. After one removal the image is 8 by 7. You recompute scores. Closing the gap changed the neighbors, so the old path is not a path on the narrower image.

Scale shrinks everything. A seam removes one cheap path and can spare a marked face.

Uniform scale

Face and wall both shrink

No path of 8 pixels

The width change is global

One seam

One pixel from each row

Remove the cheap path

Penalty 100 on the face

From 8 columns to 7

1. Each candidate seam has 8 pixels, one per row, and its cost is the sum of those scores.

2. You add 100 to the scores in columns 3, 4, and 5 so a path through the face is expensive.

3. You delete the cheapest seam. The image is now 8 by 7, and you recompute scores before any second seam.

Problem 1. How many pixels does one seam remove from an image that is 8 pixels tall?

Answer. 8, one from each row. The width drops by 1, from 8 columns to 7.

Problem 2. Why recompute scores after a removal instead of deleting the same seam again?

Answer. Closing the gap changes the neighbors of the remaining pixels, so the scores change. The old path is not a path in the 8 by 7 image.

Lecture 7 · 90 minutes

Copy one pixel, not the tile

You synthesize texture by copying pixels whose neighborhoods match a sample, not by stretching it. The sample is 4 by 4. The output is 8 by 8, which you do not have. For each new pixel you find the sample site whose neighborhood is closest to the neighbors already filled. You copy that 1 pixel. Copying the whole 4 by 4 would tile the output and show a grid seam.

You seed the output so the first comparison has neighbors. Copy the top-left 2 by 2 of the sample, 4 pixels, into the corner, then grow. If two sites match equally, either pick is allowed, and a second run may differ. You do not ignore the neighborhood and repeat the 4 by 4 on a lattice. That repeat seams every 4 columns. Check columns 3 and 4 before you accept the 8 by 8.

A seed gives the first neighborhood. After that, one pixel is copied at a time.

1. Seed a 2 by 2 block

2. Match the neighborhood

3. Copy a single pixel

4. Check every 4th column

Growing past the seed

1. The sample is 4 by 4. You copy its top-left 2 by 2, four pixels, into the output as the seed.

2. The next pixel compares its filled neighbors with the sample and copies one pixel from the closest match.

3. A tiled copy would seam every 4 columns. You inspect columns 3 and 4 before you accept the 8 by 8 result.

Problem 1. Why copy one pixel from the best neighborhood match instead of pasting the whole 4 by 4 sample?

Answer. Pasting the whole sample tiles it, and a seam appears every 4 pixels. Copying one pixel lets the next neighborhood continue the texture without that grid.

Problem 2. The seed is the top-left 2 by 2 of the sample. How many pixels is that, and why do you need them?

Answer. 4 pixels. The first synthesized pixel needs neighbors to match. An empty output has no neighborhood to compare against the sample.

Lecture 8 · 90 minutes

Match the fault to the tool

You can now read a frame as an exposure of 0.01 seconds focused at 2 meters, filter it without shifting a flat gray, stitch two views from 4 point pairs, merge 3 exposures, refocus 16 rays, remove a seam of 8 pixels, and grow an 8 by 8 texture from a 4 by 4 sample. Each edit has a number you can recompute. You can also say which edit does not repair which fault.

You pick the edit from the fault. A lamp clipped at 255 needs another exposure, not a sharpen. A squeezed face needs a seam with penalty 100, not a homography. A disk from a point at 5 meters needs a light-field shift, not a 3 by 3 box, which widens the disk. You name the fault, then the mechanism, then the number that shows it ran.

The fault names the mechanism. A nearby tool can make the fault worse.

The fault

A pixel stuck at 255

A face squeezed flat

A disk at 5 meters

The mechanism

Use a shorter exposure

Seam, penalty 100

Shift the light field

Three faults, three tools

1. A pixel at 255 in the 0.16 second frame is dropped. The 0.01 second frame supplies the sample you divide by time.

2. A seam of 8 pixels, with a penalty of 100 on the face columns, changes the width without scaling the face.

3. A shift of 1 pixel among the 4 views refocuses. A 3 by 3 box is the wrong tool for the disk at 5 meters.

Problem 1. Combine the exposure merge and the sharpen. The long frame reads 255 and the short frame reads 20 at 0.01 seconds. What do you use, and what do you refuse to sharpen?

Answer. You use 20 divided by 0.01, and you drop the 255. You refuse to sharpen the clipped 255, because there is no detail above the clip. After the tone map you may sharpen an in-range pixel, such as 80 against a blur of 50, which becomes 95.

Problem 2. Combine the panorama blend and the seam. The overlap is 40 pixels, and an 8-pixel-tall image must lose 1 column. What is the incoming weight 10 pixels into the overlap, and how many pixels does the seam remove?

Answer. The incoming weight is 0.25, which is 10 divided by 40, and the other view gets 0.75. The seam removes 8 pixels, one per row. The blend does not change the width. The seam does. You do not use a uniform scale for that width change.